

RIX2024 - Batch 1 - Jenkins Practical Assessment
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W3H Analysis

Project Title: Multi-language Translation Microservice	
What?	How?
1. What are the necessary modules? A: Modules Needed: 1) Login and Registration. 2) User Dashboard.	Login and Registration Module: 1) Register User: Method 1: Use Simple Form Page. Method 2: Use a Modular Form Page. Method 3: Use Google OAuth and other 3rd party services.
2. What are the necessary functionalities of Login and Registration Modules? A: Login and Registration Module: 1) Register User. 2) Login as User.	2) Login as User: Method 1: Use Simple Form. Method 2: Use a Modular Form. Method 3: Use 3rd party services.
3. What are the necessary functionalities of User Dashboard? A: User Dashboard: 1) Input string to translate and perform translations. 2) Filter Previous Translations.	User Dashboard: 1) Input string to translate and perform translations: Method 1: Use Simple Forms and Buttons to perform CRUD. Method 2: Have separate pages for each operation Method 3: Use Modals.

4. What are the tables needed?

A: Tables Required:

- Translations
- Users

5. What are the fields of user table?

A: Fields of Users table:

- Id (Primary Key) - int
- Name – string
- Contact – int
- Password – string

6. What are the fields of the Translations table?

A: Fields of Expenses Table:

- Id (Primary Key) - int
- TranslateString – string
- FromLang – string
- ToLang - string

7. What are the must have dependencies?

A: Dependencies:

- axios
- react-router-dom

8. What are the validations in the application?

A: Overall validations list:

- Frontend simple JS

2) Filter Previous Translations:

Method 1: Use a simple search bar.

Method 2: Have tables and sorting buttons.

Method 3: Have a separate page to pick out exact results.

Validations. • Server Side validations. • Form Validations.	
Login and Registration Module: 1) Register User: Method 1: Use Simple Form Page. A: Since the overall scale of the application is small and data needed is rather minimal, it's wise to stick with traditional forms. 2) Login as User: Method 1: Use Simple Form. A: Since the overall scale of the application is small and data needed is rather minimal, it's wise to stick with traditional forms. User Dashboard: 1) Input string to translate and perform translations: Method 1: Use Simple Forms and Buttons to perform CRUD. A: Since the overall scale of the application is small and data needed is rather minimal, it's wise to stick with traditional forms. 2) Filter Previous Translations: Method 1: Use a simple search bar.	Login and Registration Module: 1) Register User: Method 2: Use a Modular Form Page. A: Increases the complexity of the app, and also the fact that the number of needed inputs is rather minimal. Method 3: Use Google OAuth and other 3rd party services. A: Valid method but due to the scale of the application, this method is seemingly overkill. 2) Login as User: Method 2: Use a Modular Form. A: Increases the complexity of the app, and also the fact that the number of needed inputs is rather minimal. Method 3: Use 3rd party services. A: Valid method but due to the scale of the application, this method is seemingly overkill. User Dashboard:

A: Effective as it may, the simple input field can be directly mapped with the Expenses that are being displayed.	1) Input string to translate and perform translations: Method 2: Have separate pages for each operation. A: Since the admin needs all information in one place and also that the number of operations they can do is limited, it is best to have a single page control centre type. 2) Filter Previous Translations: Method 2: Have tables and sorting buttons. A: This method is also viable but not selected due to the slight increase in the number of code needed for its execution. Method 3: Have a separate page to pick out exact results. A: Having the all the data displayed in one page and search functionality in another is not ideal. Hence, this method is rejected.
Why?	Why Not?

Output Screenshots

Jenkins Initial Run

```
C:\Windows\System32\cmd.exe X + v
Microsoft Windows [Version 10.0.26100.4946]
(c) Microsoft Corporation. All rights reserved.

D:\Jenkins>java -jar jenkins.war
Running from: D:\Jenkins\jenkins.war
webroot: C:\Users\nisanth.saravanan\.jenkins\war
2025-08-26 09:40:24.873+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
```

```
*****
*****
*****

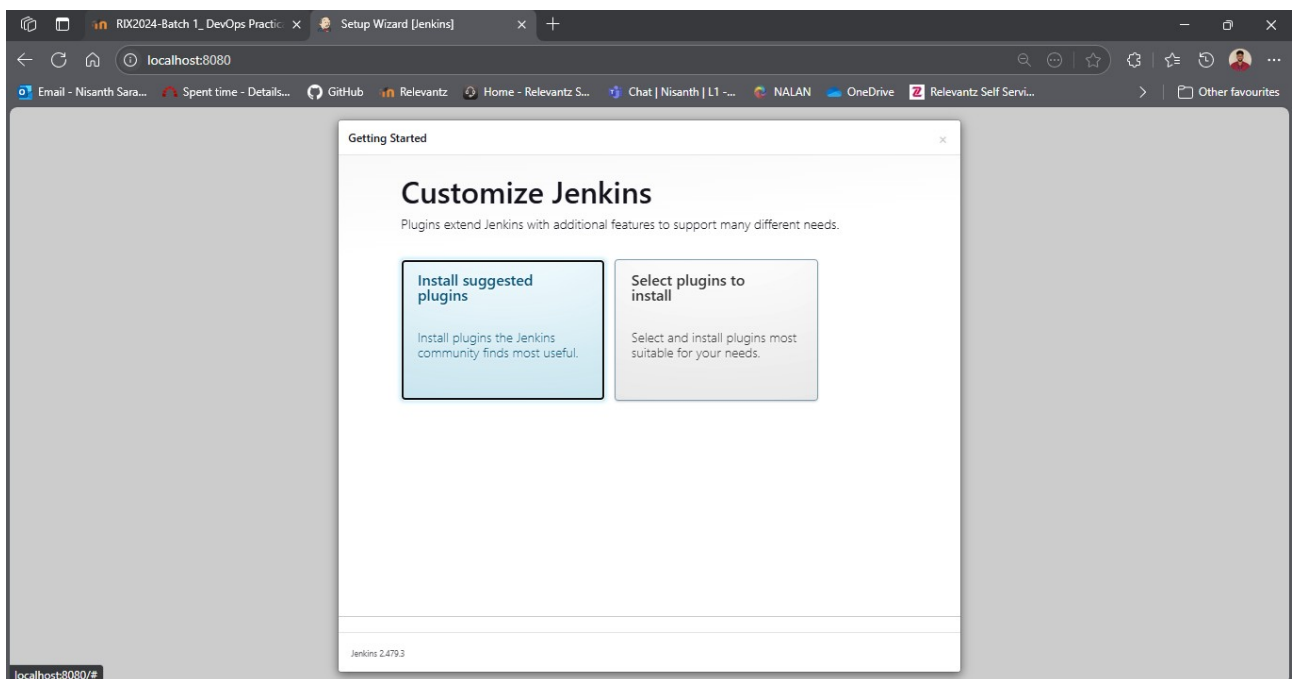
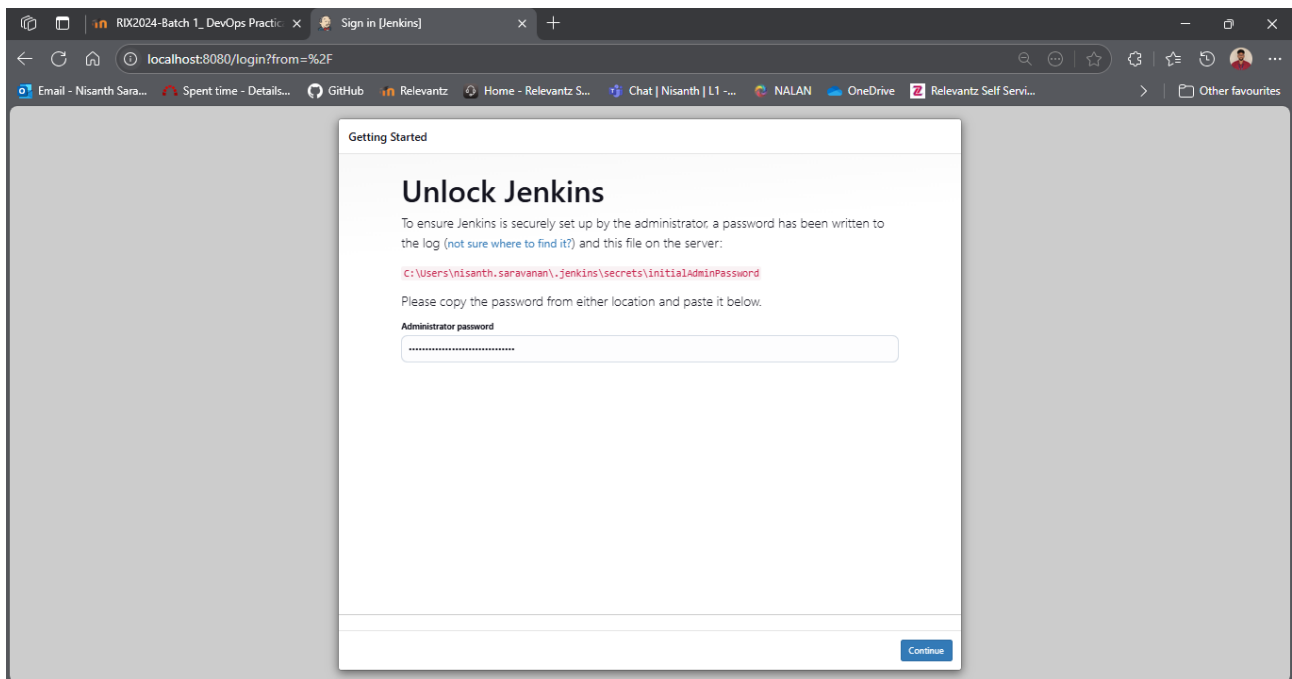
Jenkins initial setup is required. An admin user has been created and a password generated.
Please use the following password to proceed to installation:

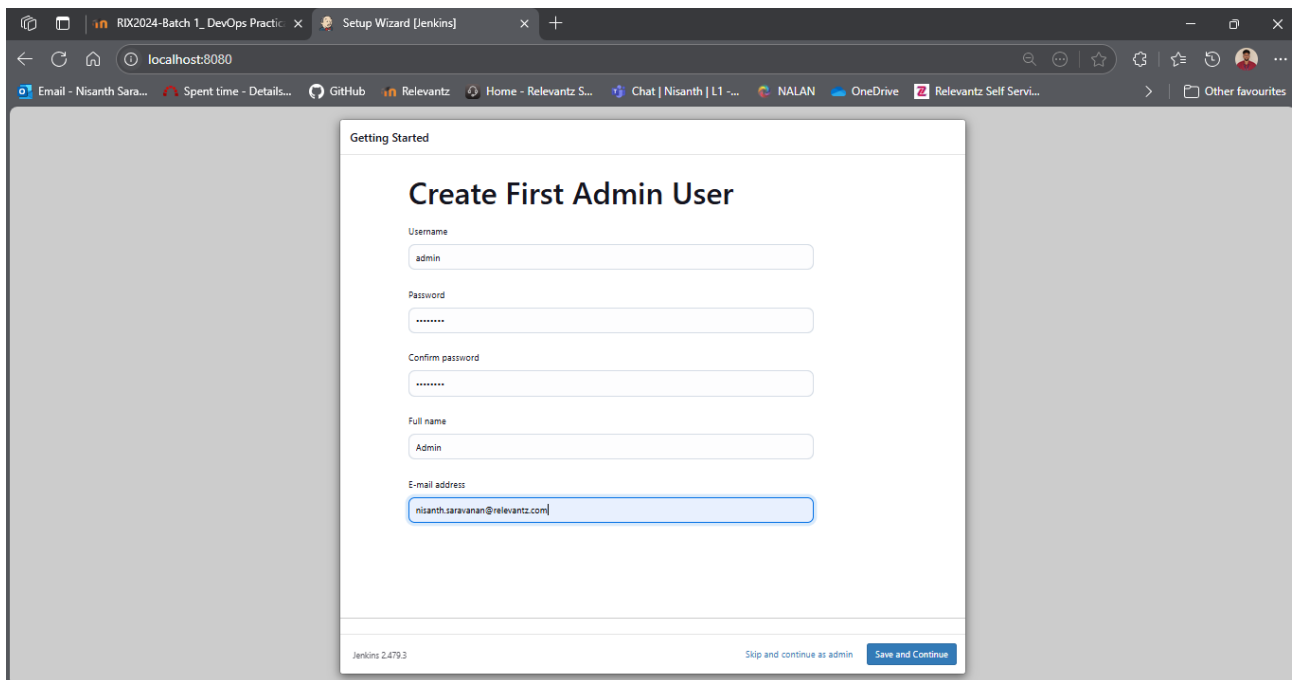
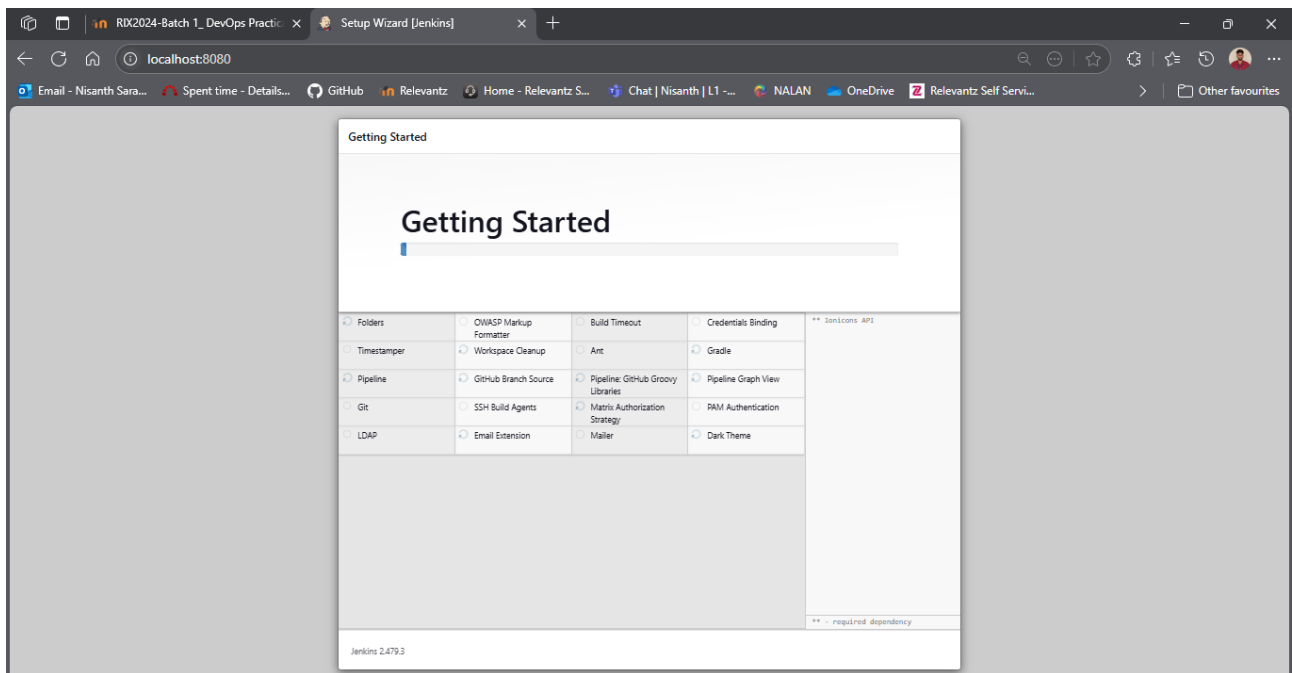
3959c36c2d4c4ec5943cd79db2274bca

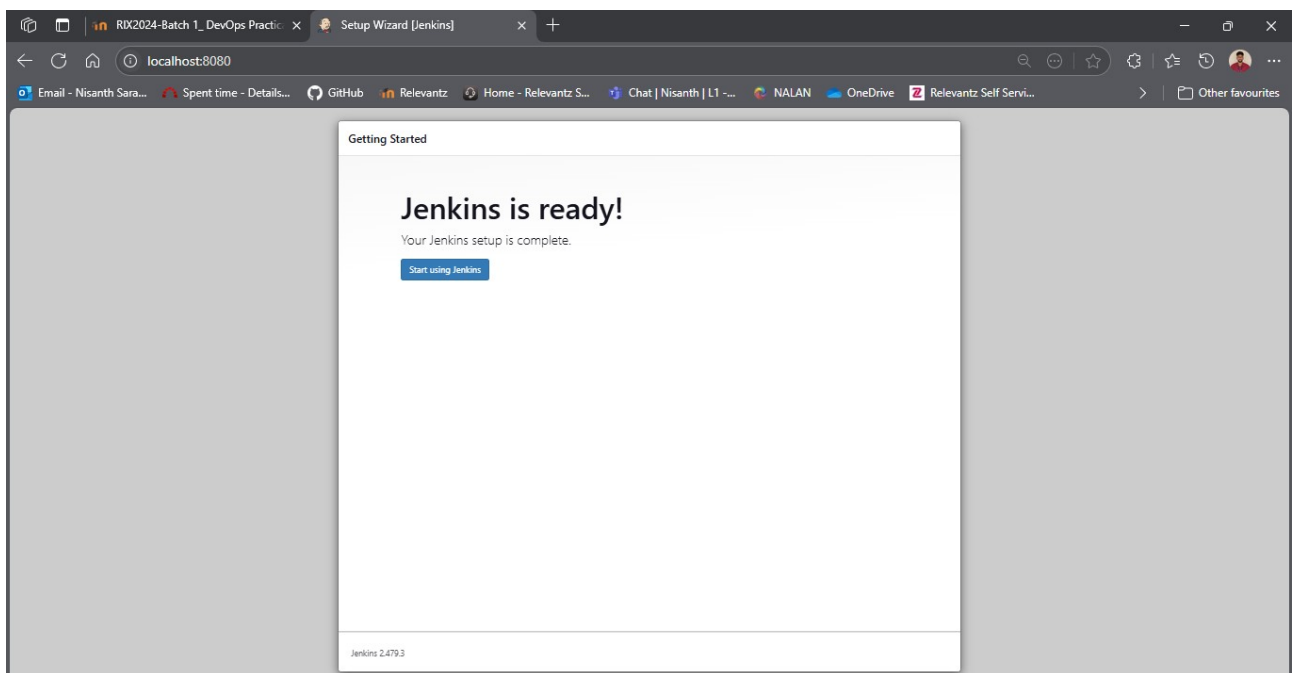
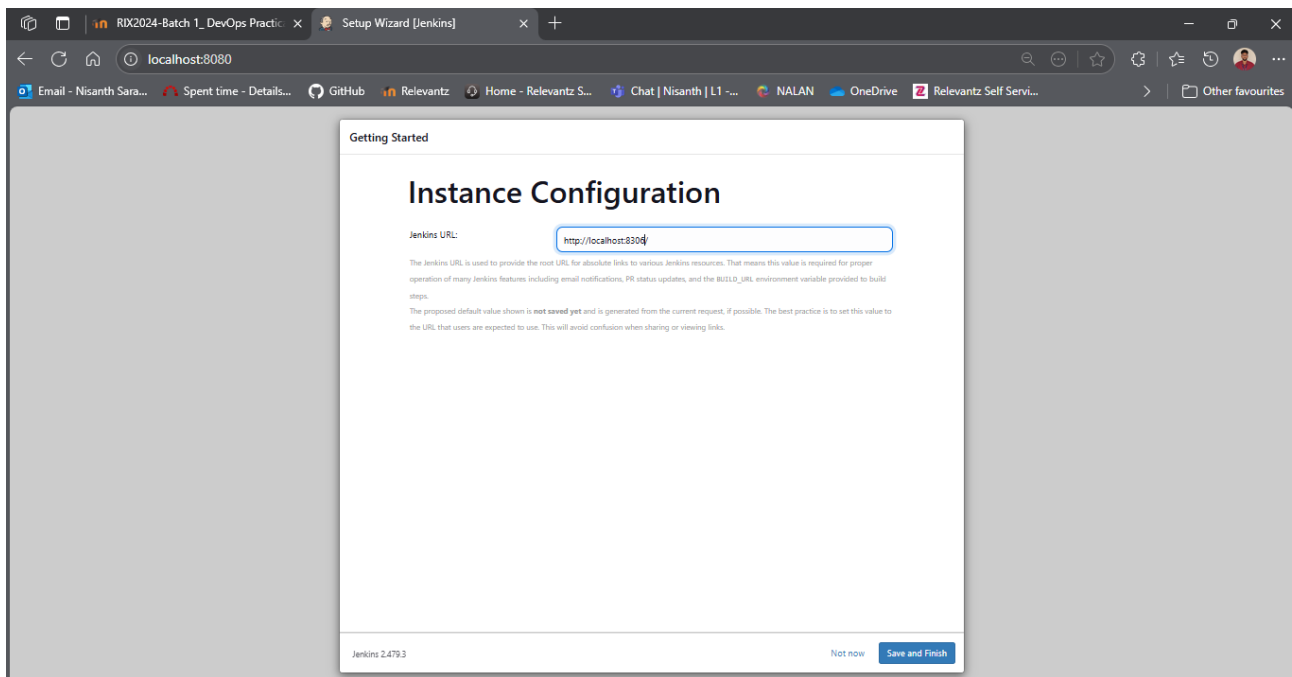
This may also be found at: C:\Users\nisanth.saravanan\.jenkins\secrets\initialAdminPassword

*****
*****
*****
```

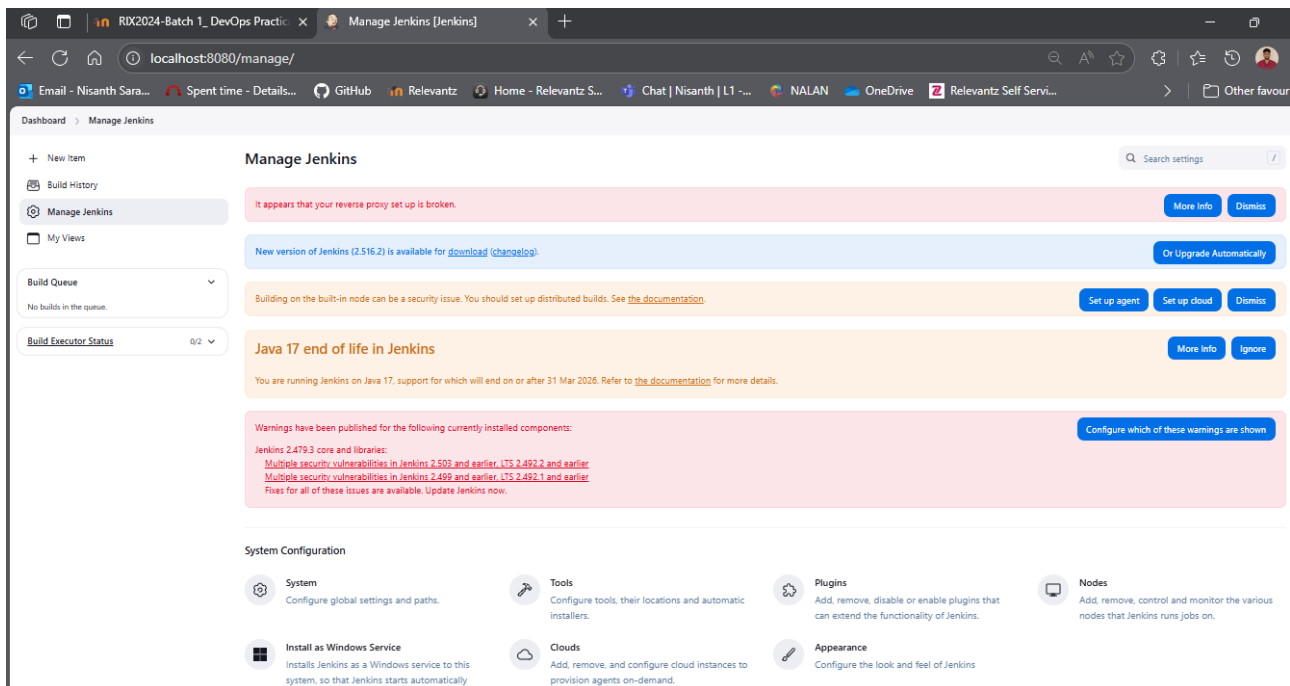
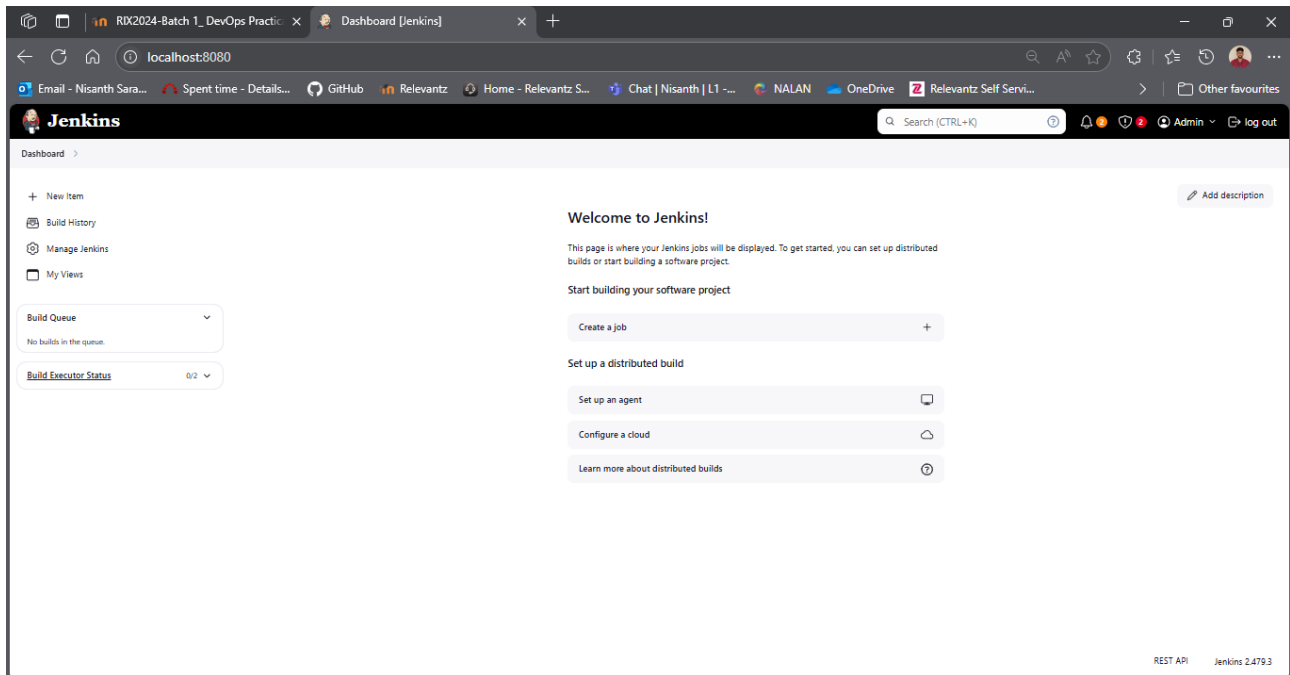
```
2025-08-26 09:41:11.813+0000 [id=33] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
2025-08-26 09:41:11.872+0000 [id=26] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2025-08-26 09:41:14.263+0000 [id=69] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenIn
staller
2025-08-26 09:41:14.263+0000 [id=69] INFO hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
```







Configuring Jenkins



Plug-ins

Dashboard

Manage Jenkins

Plugins

Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

nunit

Released

Install

✓

Docker

1274.vc0203f9f2e74

Cloud Providers

Cluster Management

docker

5 mo 21 days ago

This plugin integrates Jenkins with Docker

✓

Docker Pipeline

621.va_73f881d9232

pipeline

DevOps

Deployment

docker

2 mo 27 days ago

Build and use Docker containers from pipelines.

✓

docker-build-step

2.12

Build Tools

docker

1 yr 3 mo ago

This plugin allows to add various docker commands to your job as build steps.

Warning: This plugin version may not be safe to use. Please review the following security notices:

• [CSRF vulnerability and missing permission check](#)

✓

Pipeline: Stage View

2.38

User Interface

3 mo 28 days ago

Pipeline Stage View Plugin.

✓

SonarQube Scanner

2.18

External Site/Tool Integrations

Build Reports

6 mo 29 days ago

This plugin allows an easy integration of SonarQube, the open source platform for Continuous Inspection of code quality.

✓

MSBuild

1.37

.NET Development

Build Tools

1 mo 20 days ago

This plugin makes it possible to build a Visual Studio project (.proj) and solution files (.sln).

✓

NUnit

571.v30a_1e3f86709

Build Reports

6 mo 4 days ago

This plugin transforms NUnit test reports so they can be recorded by Jenkins' JUnit features.

Docker Commons	✓ Success
Apache HttpComponents Client 5.x API	✓ Success
Commons Compress API	✓ Success
Docker API	✓ Success
Docker	✓ Success
Docker Pipeline	✓ Success
Javadoc	✓ Success
Dev Tools Symbols API	✓ Success
JSch dependency	✓ Success
Maven Integration	✓ Success
docker-build-step	✓ Success
Pipeline Graph Analysis	✓ Success
Pipeline: REST API	✓ Success
Pipeline: Stage View	✓ Success
SonarQube Scanner	✓ Success
MSBuild	✓ Success
NUnit	✓ Success
Loading plugin extensions	✓ Success

System:

SonarQube

SonarQube servers

If checked, job administrators will be able to inject a SonarQube server configuration as environment variables in the build.

☐ Environment variables

SonarQube installations

List of SonarQube installations

Name

SonarQube

Server URL

Default is http://localhost:9000

http://localhost:9000

Server authentication token

SonarQube authentication token. Mandatory when anonymous access is disabled.

- none -

+ Add

Advanced

Add SonarQube

Docker

Declarative Pipeline (Docker)

Docker Label ?

Docker

Docker registry URL ?

tcp://localhost:2375

Registry credentials

- none -

+ Add

Tools

Java

JDK installations

Add JDK

≡

JDK

×

Name

JDK

JAVA_HOME

C:\Program Files\Java\jdk-17

☐ Install automatically ?

Add JDK

Git

Git installations

≡

Git

×

Name

Git

Path to Git executable ?

C:\Program Files\Git\bin\git.exe

☐ Install automatically ?

Add Git ▾

MSBuild

MSBuild installations

Add MSBuild

≡

MSBuild

×

Name

MSBuild

Default parameters ?

C:\Program Files\Microsoft Visual Studio\2022\Professional\MSBuild\Current\Bin

☒ Install automatically ?

≡

Install from Microsoft

×

Version ?

Build Tools 2022 ▾

Additional Resources ?

SonarQube

SonarQube Scanner installations

Add SonarQube Scanner

☰

SonarQube Scanner

✕

Name

SonarQube

SONAR_RUNNER_HOME

D:\sonarqube-25.1.0.102122\sonarqube-25.1.0.102122

☐ Install automatically ?

Add SonarQube Scanner

Docker

Docker installations

Docker installations ^  Edited

Add Docker

☰

Docker

✕

Name

Docker

Installation root ?

C:\Program Files\Docker\Docker

☐ Install automatically ?

Add Docker

Getting Ready

Creating a new public repository

Create a new repository

Preview [Switch back to classic experience](#)

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk ().*

1

General

Owner *

BonsaiPlays

Repository name *

JenkinsAssessment

✓ JenkinsAssessment is available.

Great repository names are short and memorable. How about [legendary-goggles?](#)

Description

0 / 350 characters

Pushing the existing code into the repository

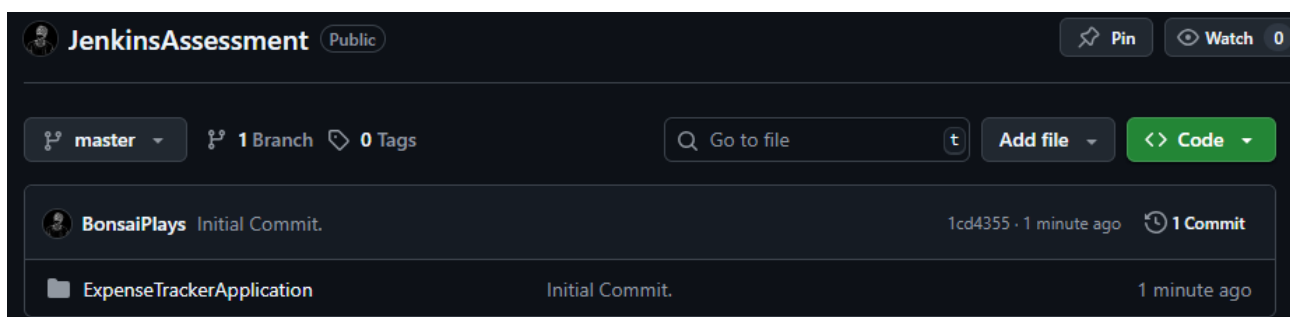
```
D:\Repos\12576_Nisanth_Saravanan_Jenkins>git init
Initialized empty Git repository in D:/Repos/12576_Nisanth_Saravanan_Jenkins/.git/

D:\Repos\12576_Nisanth_Saravanan_Jenkins>git remote add origin https://github.com/BonsaiPlays/JenkinsAssessment.git

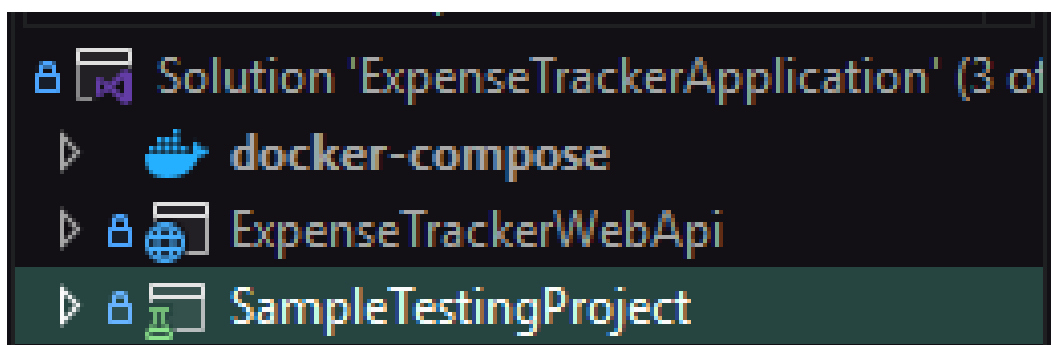
D:\Repos\12576_Nisanth_Saravanan_Jenkins>git add .
warning: in the working copy of 'ExpenseTrackerApplication/expense-management-frontend/.gitignore', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'ExpenseTrackerApplication/expense-management-frontend/README.md', LF will be replaced by CRLF the next time Git touches it
```

```
D:\Repos\12576_Nisanth_Saravanan_Jenkins>git commit -m "Initial Commit."
[master (root-commit) 1cd4355] Initial Commit.
235 files changed, 24968 insertions(+)
create mode 100644 ExpenseTrackerApplication/.dockerignore
create mode 100644 ExpenseTrackerApplication/.vs/ExpenseTrackerApplication/CopilotIndices/17.14.794.21998/CodeChunks.db
create mode 100644 ExpenseTrackerApplication/.vs/ExpenseTrackerApplication/CopilotIndices/17.14.794.21998/SemanticSymbols.db
create mode 100644 ExpenseTrackerApplication/.vs/ExpenseTrackerApplication/DesignTimeBuild/dtbcache.v2
create mode 100644 ExpenseTrackerApplication/.vs/ExpenseTrackerApplication/FileContentIndex/0285100e-74a2-44a2-a531-8b8ec5996e86.vsidx
create mode 100644 ExpenseTrackerApplication/.vs/ExpenseTrackerApplication/FileContentIndex/0ee08708-dda2-47bc-bbe9-362cbee6bc05.vsidx
create mode 100644 ExpenseTrackerApplication/.vs/ExpenseTrackerApplication/FileContentIndex/1a4e5859-a1ba-4753-94b3-3f1ac838a28.vsidx
create mode 100644 ExpenseTrackerApplication/.vs/ExpenseTrackerApplication/FileContentIndex/4e4f8056-b113-42ae-84be-284c7acde9bc.vsidx
create mode 100644 ExpenseTrackerApplication/.vs/ExpenseTrackerApplication/FileContentIndex/65f7107b-4f86-47a8-ba22-8c1fc7055273.vsidx
create mode 100644 ExpenseTrackerApplication/.vs/ExpenseTrackerApplication/v17/.fudcache.v2
create mode 100644 ExpenseTrackerApplication/.vs/ExpenseTrackerApplication/v17/.suo
create mode 100644 ExpenseTrackerApplication/.vs/ExpenseTrackerApplication/v17/DocumentLayout.backup.json
create mode 100644 ExpenseTrackerApplication/.vs/ExpenseTrackerApplication/v17/DocumentLayout.json
create mode 100644 ExpenseTrackerApplication/.vs/ProjectEvaluation/expensestrackerapplication.metadata.v9.bin
create mode 100644 ExpenseTrackerApplication/.vs/ProjectEvaluation/expensestrackerapplication.projects.v9.bin
create mode 100644 ExpenseTrackerApplication/.vs/ProjectEvaluation/expensestrackerapplication.strings.v9.bin
```

```
D:\Repos\12576_Nisanth_Saravanan_Jenkins>git push origin master
git: 'credential-manager-core' is not a git command. See 'git --help'.
Enumerating objects: 257, done.
Counting objects: 100% (257/257), done.
Delta compression using up to 12 threads
Compressing objects: 100% (237/237), done.
Writing objects: 100% (257/257), 10.47 MiB | 2.93 MiB/s, done.
Total 257 (delta 62), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (62/62), done.
To https://github.com/BonsaiPlays/JenkinsAssessment.git
* [new branch]      master -> master
```



Current working files



Along with a working react front-end.

Jenkins Work Flow

Create a new pipeline job in Jenkins

New Item

Enter an item name

ExpenseManager

Select an item type



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.



Pipeline

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

Proceed with scripting

Cherry-pick code from the repository

```
stage("Git Checkout") {  
    git "https://github.com/BonsaiPlays/JenkinsAssessment.git"  
}
```

Build the Project

```
stage("Build Project") {  
    bat "dotnet build ExpenseTrackerApplication/ExpenseTrackerApplication.sln"  
}
```

Test the project

```
stage("Test Project") {  
    bat "dotnet test ExpenseTrackerApplication/ExpenseTrackerApplication.sln"  
}
```

SonarQube

```
stage("QA") {  
  withSonarQubeEnv('SonarQube') {  
    catchError(buildResult: 'SUCCESS', stageResult: 'FAILURE') {  
      bat 'dotnet sonarscanner begin /k:"ExpenseTracker" /d:sonar.host.url="http://localhost:9000" /d:sonar.token="sqp_c86a9e4e765fdd0d95012f705cc1f9492db7de99"'  
      bat 'dotnet build'  
      bat 'dotnet sonarscanner end /d:sonar.token="sqp_c86a9e4e765fdd0d95012f705cc1f9492db7de99"'  
    }  
  }  
}
```

Allows the pipeline not to be aborted due to quality gates failure, for this assessment.

Docker

```
stage("Docker Containerization") {  
  dir('ExpenseTrackerApplication'){  
    bat 'docker-compose up --build'  
  }  
}
```

use: docker-compose up -d --build
in order to run in detached mode and reach a success status in the pipeline.

Complete Pipeline Script

```
node {
    stage("Git Checkout") {
        git "https://github.com/BonsaiPlays/JenkinsAssessment.git"
    }

    stage("Build Project") {
        bat "dotnet build ExpenseTrackerApplication/ExpenseTrackerApplication.sln"
    }

    stage("Test Project") {
        bat "dotnet test ExpenseTrackerApplication/ExpenseTrackerApplication.sln"
    }

    stage("QA") {
        withSonarQubeEnv('SonarQube') {
            catchError(buildResult: 'SUCCESS', stageResult: 'FAILURE') {
                bat 'dotnet sonarscanner begin /k:"ExpenseTracker"
/d:sonar.host.url="http://localhost:9000"
/d:sonar.token="sqp_c86a9e4e765fdd0d95012f705cc1f9492db7de99"'
                bat 'dotnet build'
                bat 'dotnet sonarscanner end
/d:sonar.token="sqp_c86a9e4e765fdd0d95012f705cc1f9492db7de99"'
            }
        }
    }

    stage("Docker Containerization") {
        dir('ExpenseTrackerApplication'){
            bat 'docker-compose up -d --build'
        }
    }
}
```

Stage View Output

**Jenkins**

Dashboard > ExpenseManager >

Status

</> Changes

▶ Build Now

⚙️ Configure

🗑️ Delete Pipeline

🔍 Full Stage View

📶 SonarQube

📁 Stages

✎ Rename

❓ Pipeline Syntax

Builds

Filter

Today

✔️ #11 16:40

✔️ #5 16:25

✔️ #3 16:22

✔️ #2 16:21

✔️ #1 16:20

ExpenseManager

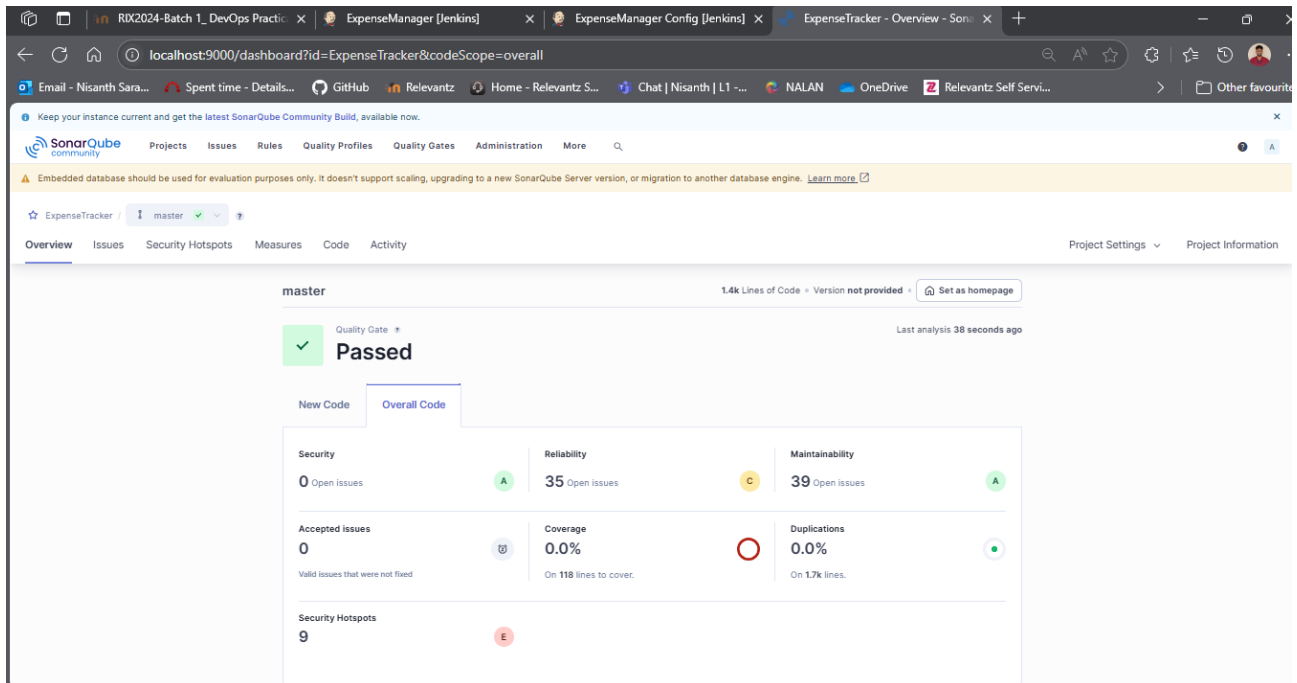
Stage View

Average stage times:
(full run time: ~47s)

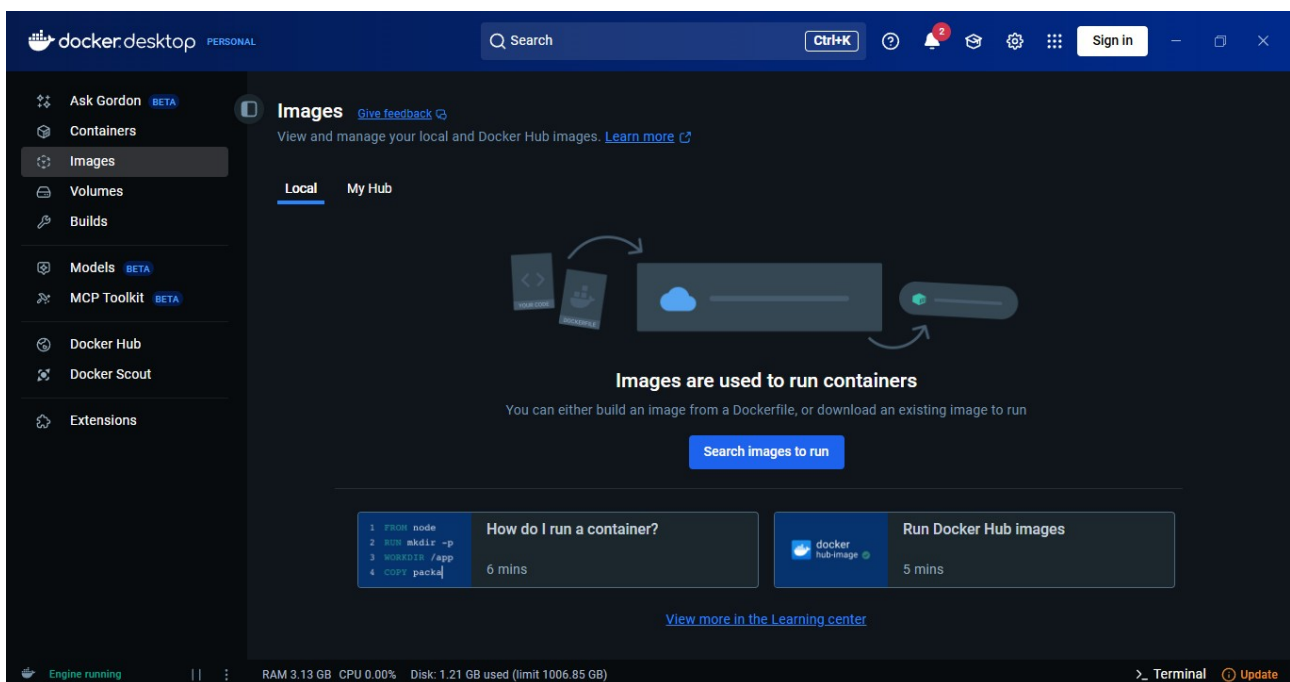
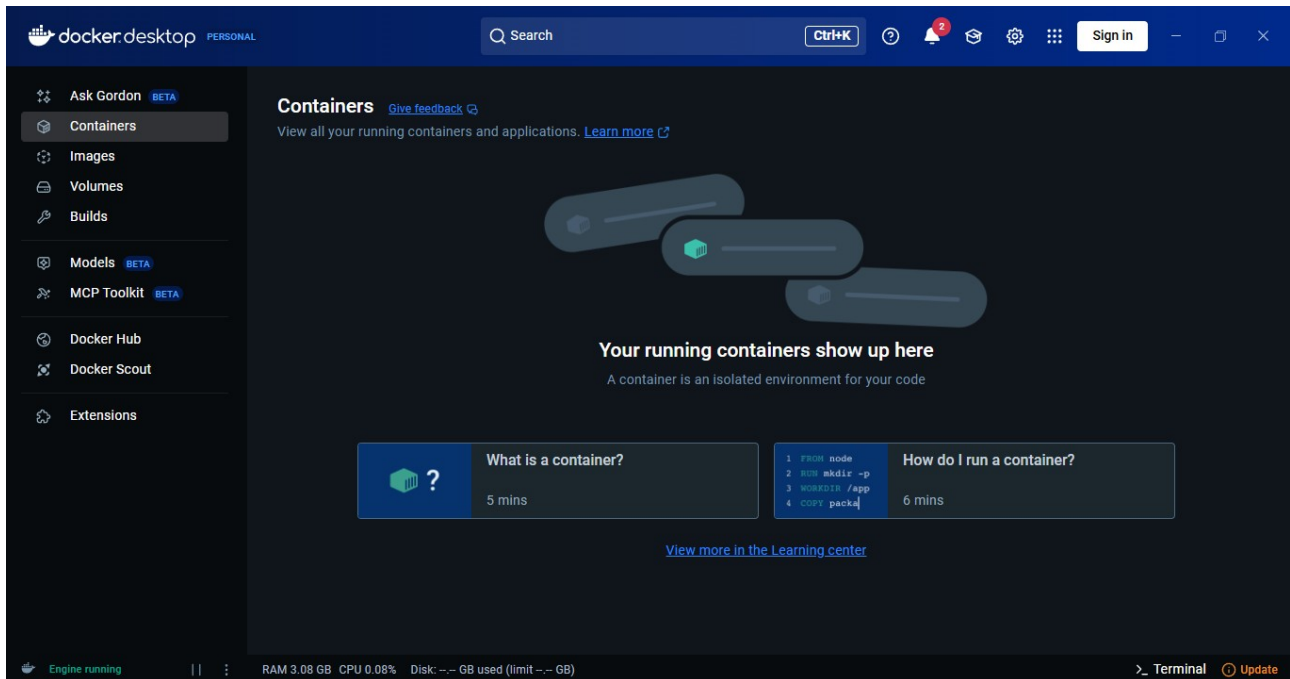
	Git Checkout	Build Project	Test Project	QA	Docker Containerization
#11 Aug 26 16:40 No Changes	4s	15s	8s	40s	15s
#5 Aug 26 16:25 No Changes	6s	16s	8s	37s	
#3 Aug 26 16:22 No Changes	7s	10s	6s		
#2 Aug 26 16:21 No Changes	7s	24s			
#1 Aug 26 16:20 No Changes	23s				

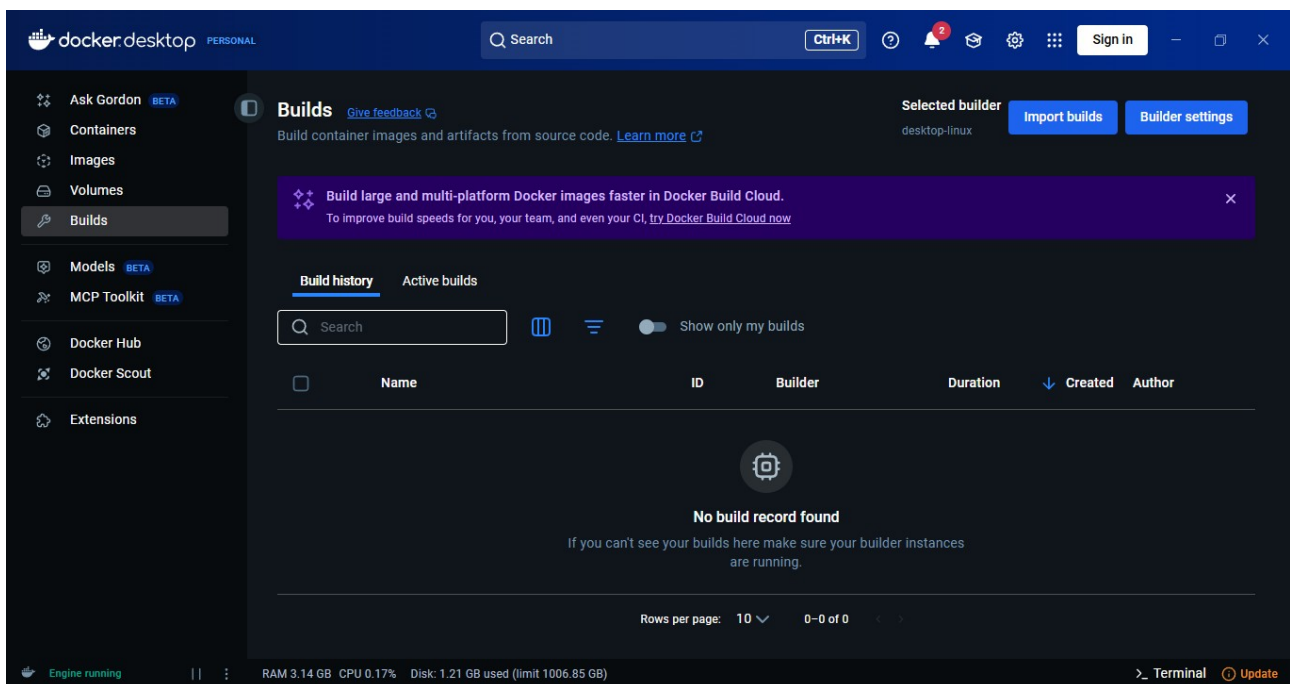
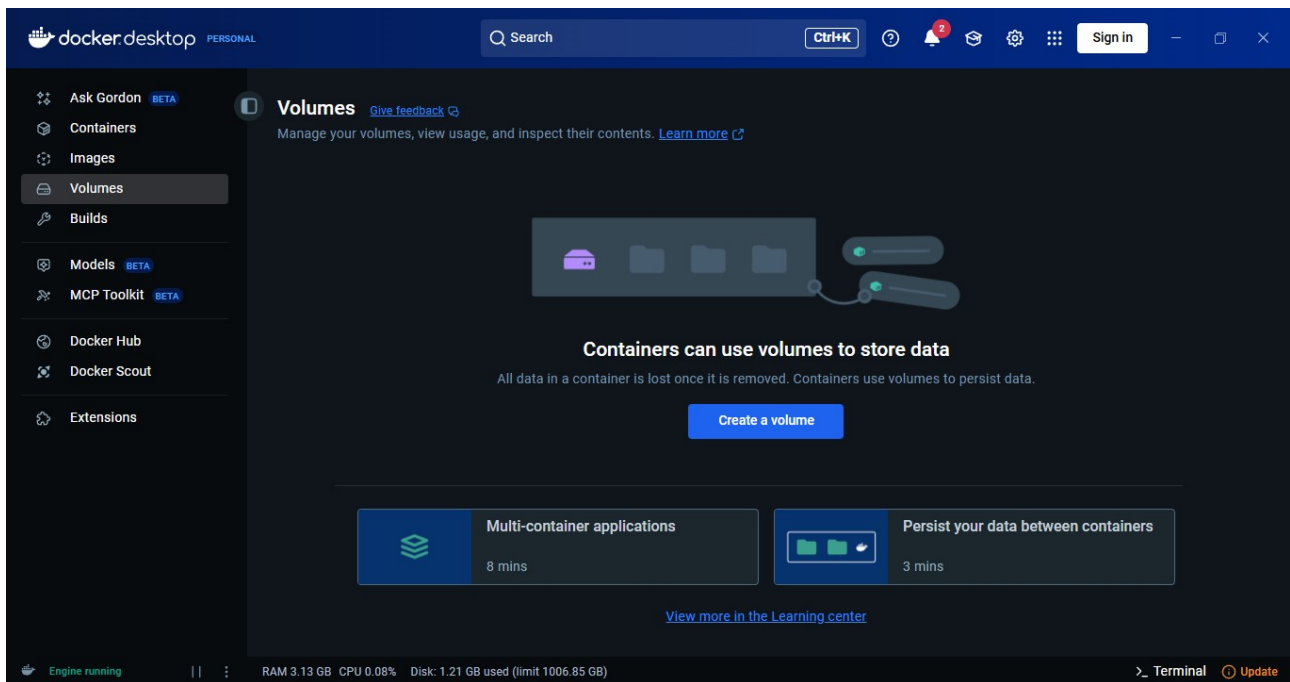
Permalinks

SonarQube Output

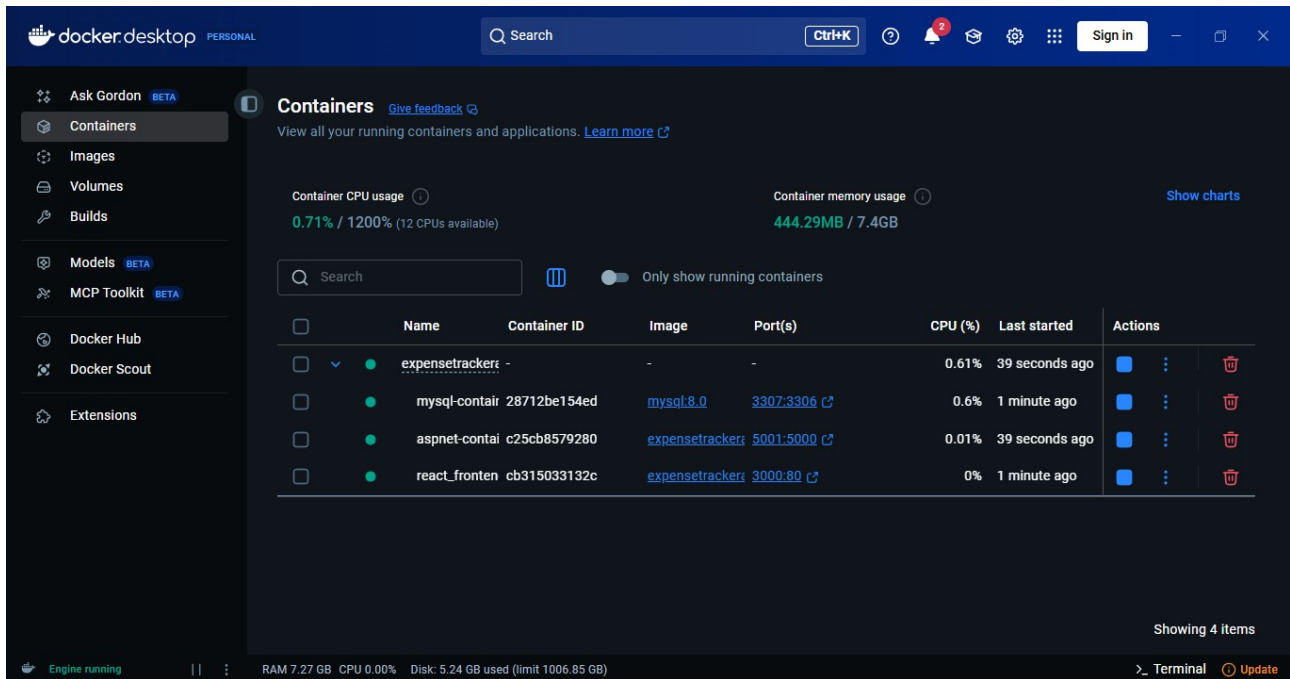


Docker Desktop Output – Initial





Docker Desktop After a Successful Jenkins Build Run



The screenshot shows the Docker Desktop interface with the 'Containers' tab selected. The left sidebar contains navigation options: Ask Gordon, Containers, Images, Volumes, Builds, Models, MCP Toolkit, Docker Hub, Docker Scout, and Extensions. The main panel displays container statistics and a list of running containers.

Containers [Give feedback](#)

View all your running containers and applications. [Learn more](#)

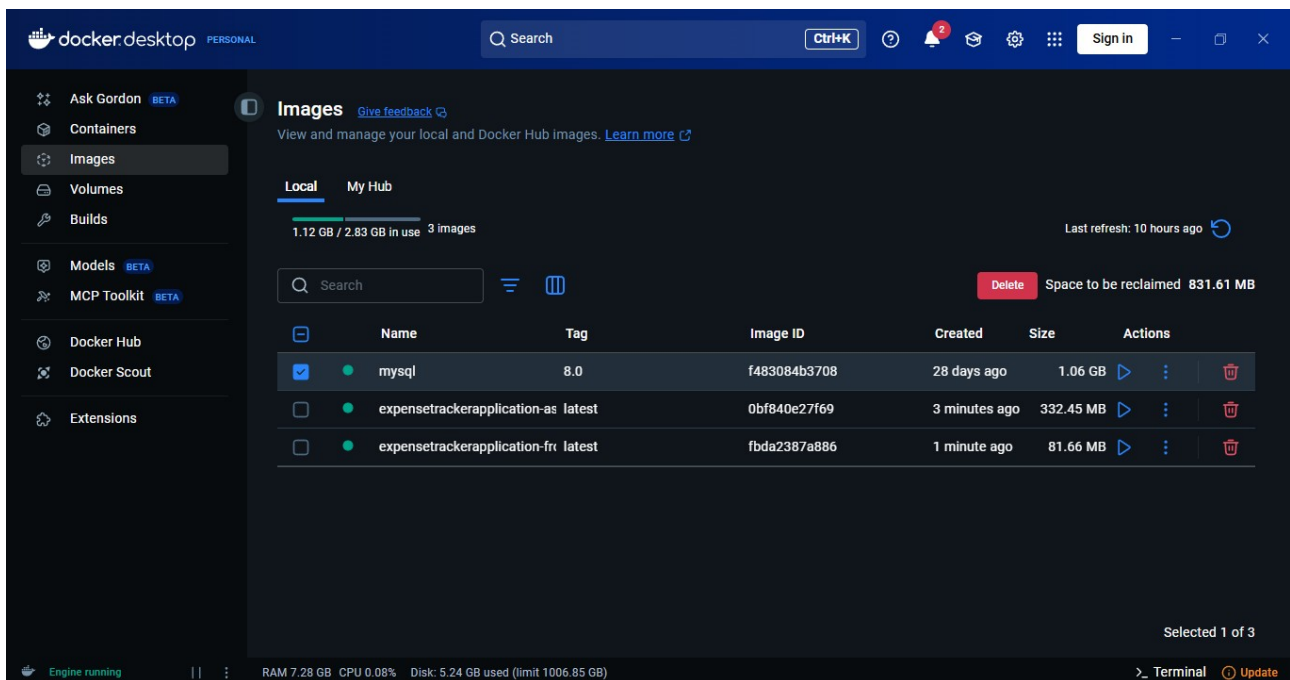
Container CPU usage 0.71% / 1200% (12 CPUs available) Container memory usage 444.29MB / 7.4GB [Show charts](#)

Search Only show running containers

	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	expensetracker	-	-	-	0.61%	39 seconds ago	Stop Refresh Delete
☐	mysql-contair	28712be154ed	mysql:8.0	3307:3306	0.6%	1 minute ago	Stop Refresh Delete
☐	aspnet-contair	c25cb8579280	expensetracker	5001:5000	0.01%	39 seconds ago	Stop Refresh Delete
☐	react_frontend	cb315033132c	expensetracker	3000:80	0%	1 minute ago	Stop Refresh Delete

Showing 4 items

Engine running RAM 7.27 GB CPU 0.00% Disk: 5.24 GB used (limit 1006.85 GB) [Terminal](#) [Update](#)



The screenshot shows the Docker Desktop interface with the 'Images' tab selected. The left sidebar is the same as the previous screenshot. The main panel displays image statistics and a list of local images.

Images [Give feedback](#)

View and manage your local and Docker Hub images. [Learn more](#)

Local My Hub

1.12 GB / 2.83 GB in use 3 Images Last refresh: 10 hours ago [Refresh](#)

Search [Delete](#) Space to be reclaimed 831.61 MB

	Name	Tag	Image ID	Created	Size	Actions
☒	mysql	8.0	f483084b3708	28 days ago	1.06 GB	Run Refresh Delete
☐	expensetrackerapplication-as	latest	0bf840e27f69	3 minutes ago	332.45 MB	Run Refresh Delete
☐	expensetrackerapplication-frt	latest	fbda2387a886	1 minute ago	81.66 MB	Run Refresh Delete

Selected 1 of 3

Engine running RAM 7.28 GB CPU 0.08% Disk: 5.24 GB used (limit 1006.85 GB) [Terminal](#) [Update](#)

docker:desktopPERSONAL

Q Search

Ctrl+K

2

Sign in

Ask GordonBETA

Containers

Images

Volumes

Builds

ModelsBETA

MCP ToolkitBETA

Docker Hub

Docker Scout

Extensions

Volumes

Give feedback

Manage your volumes, view usage, and inspect their contents. [Learn more](#)

Q Search

Create

	Name	Created	Size	Actions
	expensetrackerapplication_mysqlvolume	2 minutes ago	0 Bytes	

Walkthroughs

Multi-container applications
8 mins

Persist your data between containers
3 mins

[View more in the Learning center](#)

Engine running

RAM 7.30 GB CPU 0.42% Disk: 5.24 GB used (limit 1006.85 GB)

TerminalUpdate

docker:desktopPERSONAL

Q Search

Ctrl+K

2

Sign in

Ask GordonBETA

Containers

Images

Volumes

Builds

ModelsBETA

MCP ToolkitBETA

Docker Hub

Docker Scout

Extensions

Builds

Give feedback

Build container images and artifacts from source code. [Learn more](#)

Selected builder

desktop-linux

Import builds

Builder settings

Build large and multi-platform Docker images faster in Docker Build Cloud.

To improve build speeds for you, your team, and even your CI, [try Docker Build Cloud now](#)

Build history

Active builds

Q Search

Show only my builds

	Name	ID	Builder	Duration	Created	Author	
	Intern Files/Intern Assessment/L1 Assessmen...	xyu3nt	default	1m 31s	5 minutes ago	N/A	
	Intern Files/Intern Assessment/L1 Assessmen...	t74l2x	default	3m 07s	5 minutes ago	N/A	

Rows per page: 10 1-2 of 2

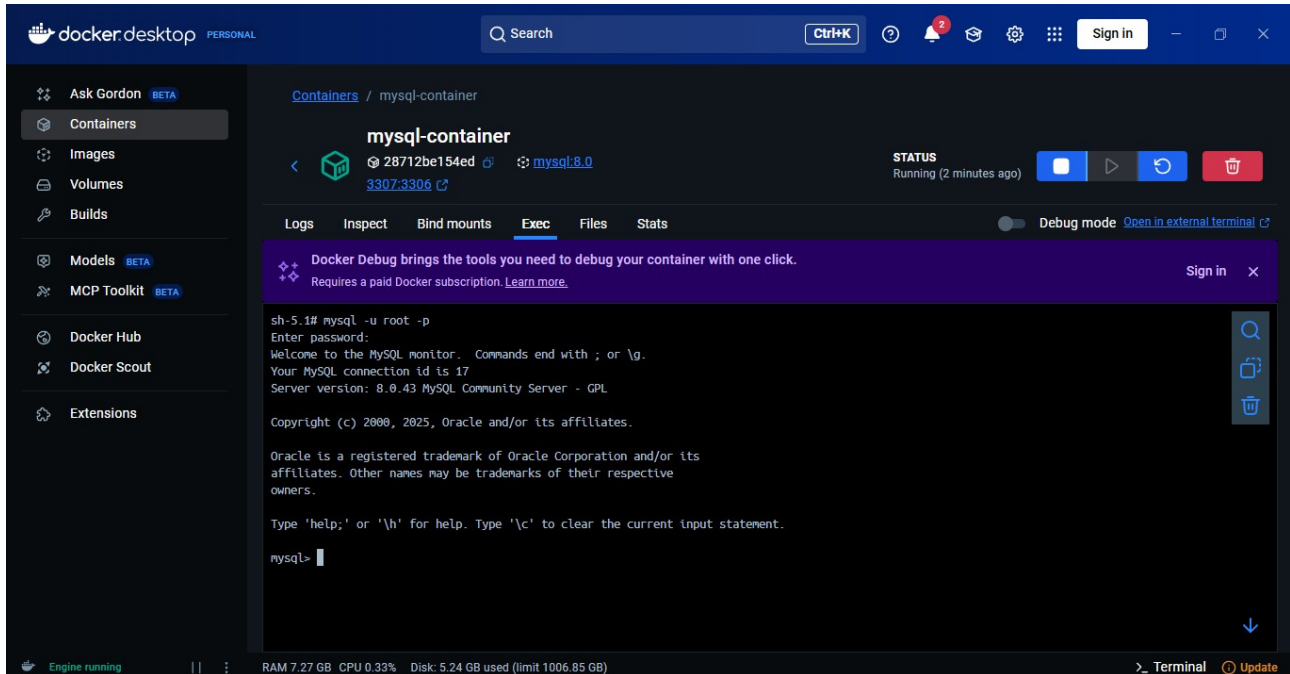
Engine running

RAM 7.30 GB CPU 0.08% Disk: 5.24 GB used (limit 1006.85 GB)

TerminalUpdate

Containerized Outputs

Database

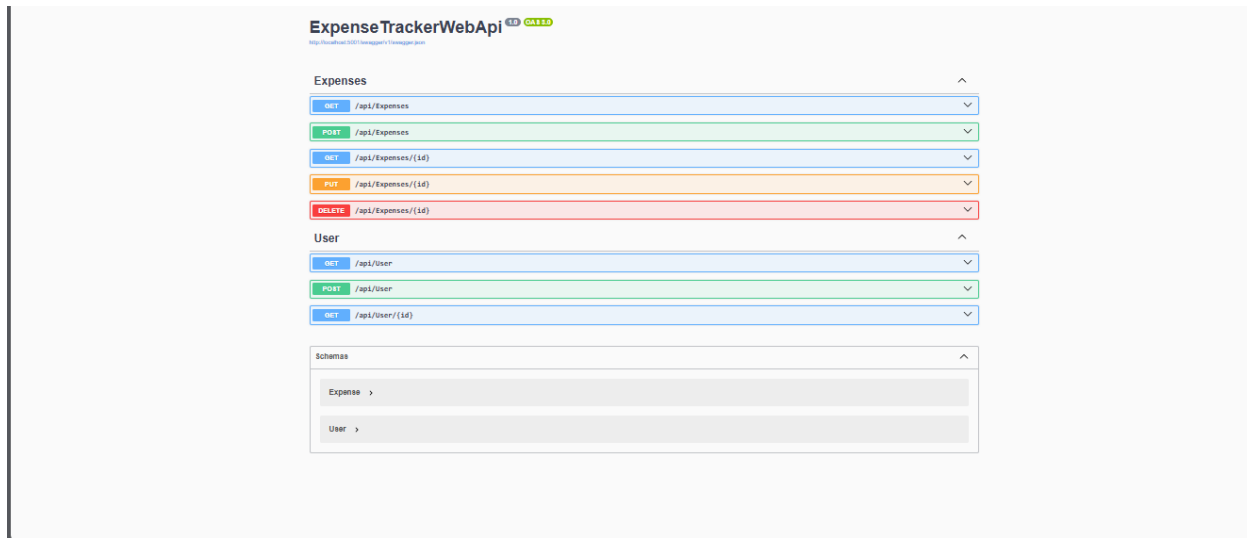


```
mysql> show databases;
+-----+
| Database |
+-----+
| ExpenseDb |
| information_schema |
| mysql |
| performance_schema |
| productdb |
| sys |
+-----+
6 rows in set (0.01 sec)
```

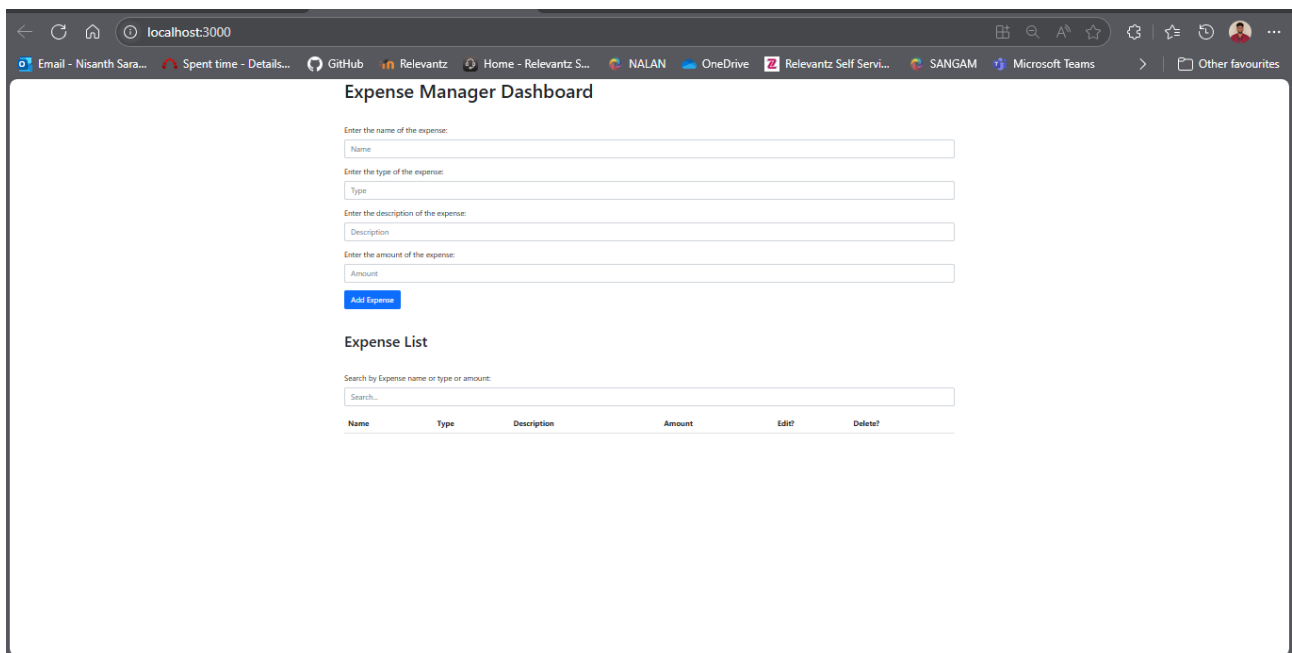
```
mysql> use ExpenseDb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> show tables;
+-----+
| Tables_in_ExpenseDb |
+-----+
| Expenses |
| Users |
| __EFMigrationsHistory |
+-----+
3 rows in set (0.00 sec)
```

Web API



React Frontend



CRUD Operations

Create - Without Description

Expense Manager Dashboard

Enter the name of the expense:

Enter the type of the expense:

Enter the description of the expense:

Enter the amount of the expense:

Expense List

Search by Expense name or type or amount:

Name	Type	Description	Amount	Edit?	Delete?
------	------	-------------	--------	-------	---------

```
mysql> use ExpenseDb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> select * from Expenses;
+-----+-----+-----+-----+
| Id | Name           | Type       | Description    | Price |
+-----+-----+-----+-----+
| 2  | Electricity Bills | Necessity | No Description. | 250000 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Create – With Description

Expense Manager Dashboard

Enter the name of the expense:

Enter the type of the expense:

Enter the description of the expense:

Enter the amount of the expense:

Add Expense

Expense List

Search by Expense name or type or amount:

Name	Type	Description	Amount	Edit?	Delete?
Electricity Bills	Necessity	No Description.	Rs.250000	<button>Edit</button>	<button>Delete</button>

```
mysql> select * from Expenses;
+----+-----+-----+-----+-----+
| Id | Name      | Type   | Description          | Price |
+----+-----+-----+-----+-----+
| 2  | Electricity Bills | Necessity | No Description.      | 250000 |
| 3  | Veggies      | Food    | Tomatoes still left on the list | 134   |
+----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Edit

Expense List

Search by Expense name or type or amount:

Search...					
Name	Type	Description	Amount	Edit?	Delete?
Electricity Bills	Necessity	No Description.	Rs.250000	<button>Edit</button>	<button>Delete</button>
Veggies	Food	Tomatoes still left on the list	Rs.134	<button>Edit</button>	<button>Delete</button>

From

Expense Manager Dashboard

Enter the name of the expense:

Enter the type of the expense:

Enter the description of the expense:

Enter the amount of the expense:

Update ExpenseCancel

To

Expense Manager Dashboard

Enter the name of the expense:

Enter the type of the expense:

Enter the description of the expense:

Enter the amount of the expense:

Update ExpenseCancel

Expense List

Search by Expense name or type or amount:

Search...					
Name	Type	Description	Amount	Edit?	Delete?
Electricity Bills	Necessity	The next month bill is 10% off.	Rs.250000	<button>Edit</button>	<button>Delete</button>
Veggies	Food	Tomatoes still left on the lst	Rs.134	<button>Edit</button>	<button>Delete</button>

From

```
mysql> select * from Expenses;
+----+-----+-----+-----+-----+
| Id | Name          | Type   | Description                               | Price |
+----+-----+-----+-----+-----+
| 2  | Electricity Bills | Necessity | No Description.                         | 250000 |
| 3  | Veggies          | Food    | Tomatoes still left on the lst         | 134    |
+----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

To

```
mysql> select * from Expenses;
+----+-----+-----+-----+-----+
| Id | Name          | Type   | Description                               | Price |
+----+-----+-----+-----+-----+
| 2  | Electricity Bills | Necessity | The next month bill is 10% off.       | 250000 |
| 3  | Veggies          | Food    | Tomatoes still left on the lst         | 134    |
+----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

Delete

Before

```
mysql> select * from Expenses;
+----+-----+-----+-----+-----+
| Id | Name          | Type   | Description                                     | Price |
+----+-----+-----+-----+-----+
| 2  | Electricity Bills | Necessity | The next month bill is 10% off. | 250000 |
| 3  | Veggies          | Food    | Tomatoes still left on the list | 134   |
+----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

Expense List

Search by Expense name or type or amount:

Name	Type	Description	Amount	Edit?	Delete?
Electricity Bills	Necessity	The next month bill is 10% off.	Rs.250000	Edit	Delete
Veggies	Food	Tomatoes still left on the list	Rs.134	Edit	Delete

After

```
mysql> select * from Expenses;
+----+-----+-----+-----+-----+
| Id | Name   | Type | Description                                     | Price |
+----+-----+-----+-----+-----+
| 3  | Veggies | Food | Tomatoes still left on the list | 134   |
+----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Expense List

Search by Expense name or type or amount:

Name	Type	Description	Amount	Edit?	Delete?
Veggies	Food	Tomatoes still left on the list	Rs.134	Edit	Delete

Searching

Sample Data

Expense List

Search by Expense name or type or amount:

Name	Type	Description	Amount	Edit?	Delete?
Veggies	Food	Tomatoes still left on the list	Rs.134	Edit	Delete
Electricity Bills	Necessity	No Description.	Rs.250000	Edit	Delete
Water Bills	Necessity	No Description.	Rs.250000	Edit	Delete
School Fee	Education	No Description.	Rs.250000	Edit	Delete

Searching By Name

Expense List

Search by Expense name or type or description:

Name	Type	Description	Amount	Edit?	Delete?
Electricity Bills	Necessity	No Description.	Rs.250000	Edit	Delete
Water Bills	Necessity	No Description.	Rs.250000	Edit	Delete

Searching By Type

Expense List

Search by Expense name or type or description:

Name	Type	Description	Amount	Edit?	Delete?
Veggies	Food	Tomatoes still left on the list	Rs.134	Edit	Delete

Expense List

Search by Expense name or type or description:

Name	Type	Description	Amount	Edit?	Delete?
School Fee	Education	No Description.	Rs.250000	Edit	Delete

Searching By Description

Expense List

Search by Expense name or type or description:

Name	Type	Description	Amount	Edit?	Delete?
Electricity Bills	Necessity	No Description.	Rs.250000	Edit	Delete
Water Bills	Necessity	No Description.	Rs.250000	Edit	Delete
School Fee	Education	No Description.	Rs.250000	Edit	Delete

Expense List

Search by Expense name or type or description:

Name	Type	Description	Amount	Edit?	Delete?
Veggies	Food	Tomatoes still left on the list	Rs.134	Edit	Delete